

JM20330 Serial ATA Bridge Chip

Description

JMicron's JM20330 is a single chip solution for serial and parallel ATA translation. It includes the Serial ATA PHY, Link, Transport, and Parallel ATA (application layer) controller. The Serial ATA physical, link, and transport layers are compliant to Serial ATA 1.0. JM20330 supports a 1.5GHz data rate, and scalable to 3.0 GHz data rate by directly doubling the internal clock source. The application layer supports both the ATA register command set and PACKET command set, which could drive both Hard Disk Drive and ATAPI Optical Storage such as CR-ROM, CD-RW, DVD-ROM, DVD-RW, etc. The Serial ATA and the Parallel ATA application layer support both host and device operation and can be configured through a single pin.

Features

Serial ATA 1.0 Specification compliant
Automatic Serial ATA 3.0/1.5 Gbps speed negotiation
ATA/ATAPI PIO mode 0 to 4
ATA/ATAPI Ultra DMA of transfer rate 16.7, 25, 33, 48, 66, 100, 133, and 150MB/s.
ATA/ATAPI LBA48 addressing mode associated with 2-byte sector count
Support Serial ATA hot-plug
Ultra low power consumption
Work for both AC and DC couple between the transmitter and the receiver
Provide specified OOB signal detection and transmission
Support Spread Spectrum Clocking to reduce EMI
Support 20MHz, 25MHz, 30MHz or 40MHz Reference Clock
Support Partial/Slumber power management
Provide adjustable TX signal amplitude and pre-emphasis level
Master/Slave support

Quick Reference

Signal Bit Rate	3.0/1.5 Gbps
Spread Spectrum	-3.0% to 0.0
Power Supply	3.3V and 1.8V
ESD Protection	2000 V
Host and Device	Programmable
Adj. Amplitude	2 levels
Adj. Pre-emphasis	4 levels
Package	64-pin TQFP
Applications	All SATA products Mass storage devices Optical storage Dongle bridge Storage system

Functional Block Diagram

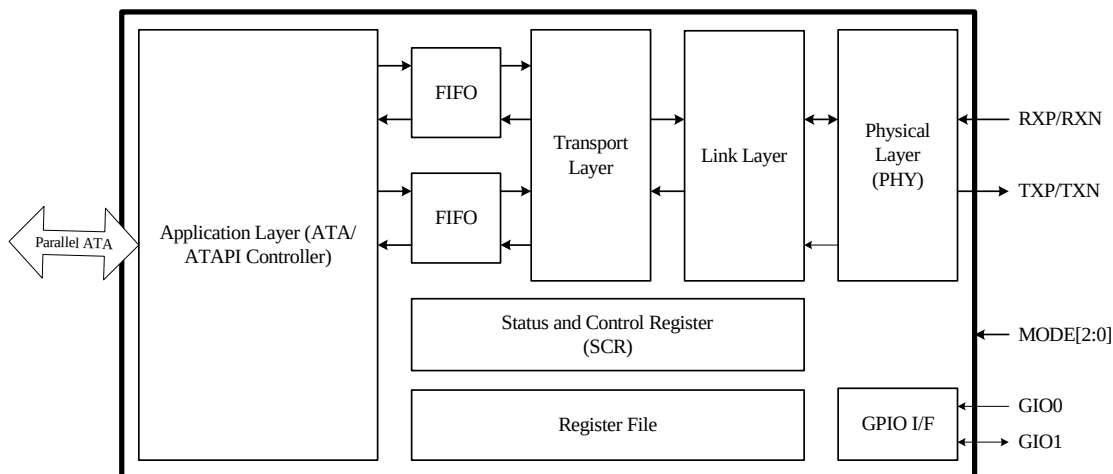


Fig. 1 Functional Block Diagram of JM20330

Applications

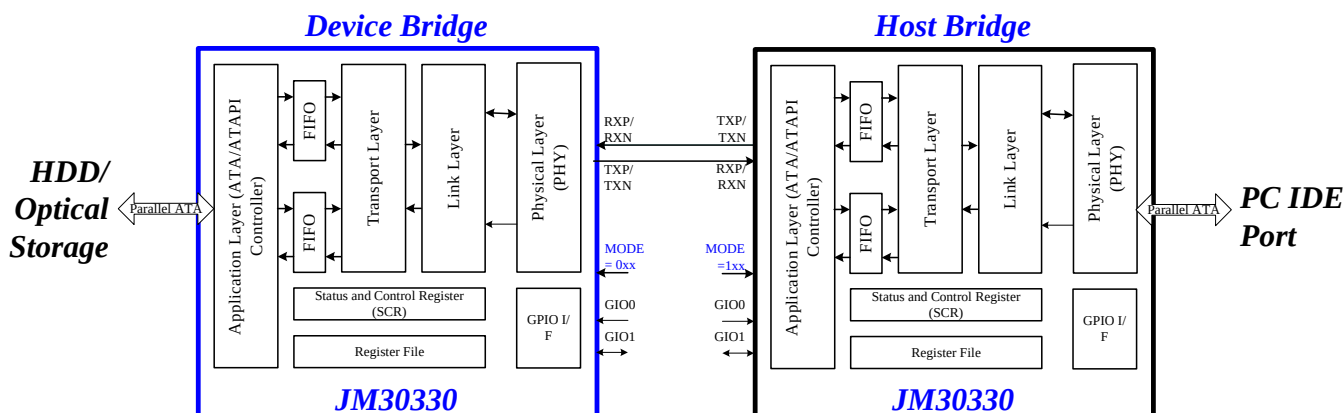


Fig. 2 JM20330 Host and Device bridge system diagram

Product Information

Name	Description
JM20330	Serial ATA Bridge Chip

Design Kit

1	JM20330 Data Sheet
2	JM20330 Design Guide
3	Application EVB

Contact Information

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